The Gradient x Input (GxI) visualizations, the mode clearly reacts to emotionally strong words, but it doesn't always do so in a consistent or human-like way.

As GxI also reveals the direction of each token's contribution, not its emotional polarity. Green bars indicate words that increase the model's confidence in the predicted emotion, where as red bars show words that lower the confidence. This can be clearly seen in fig 1, even thought "safe" is generally seen as a positive word to humans, it appears red on the GxI plot, meaning that the word reduced the models confidence in the predicted emotion (anger).

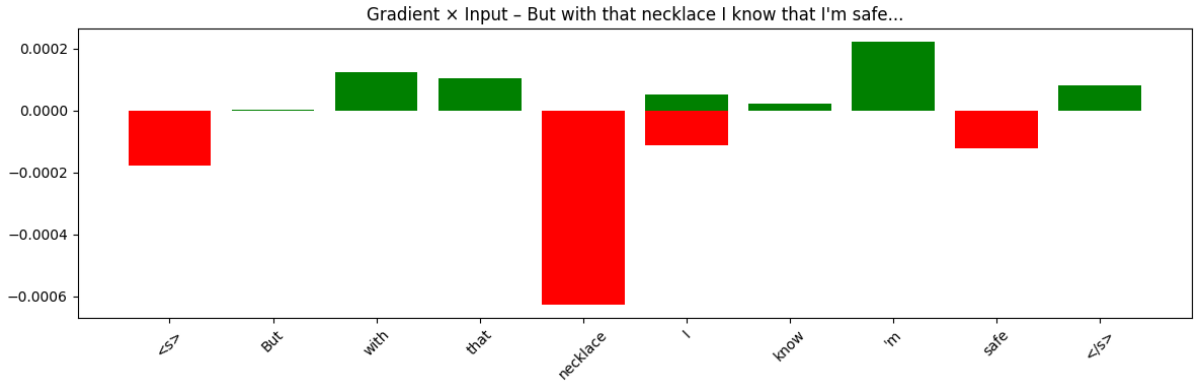


Figure 1 shows the GxI plot of the sentence "But with that necklace I know that I'm safe"

To dive further in, as seen in the fig 2 and fig 3, in sentences like "its making me feel sick" or "doing what he wants and manipulating people and I haven't stood for it a moment", the bars for tokens such as "feel sick" and "manipulating" showed the highest positive relevance, which makes sense and aligns with what we expect since those words convey a strong negative emotion.

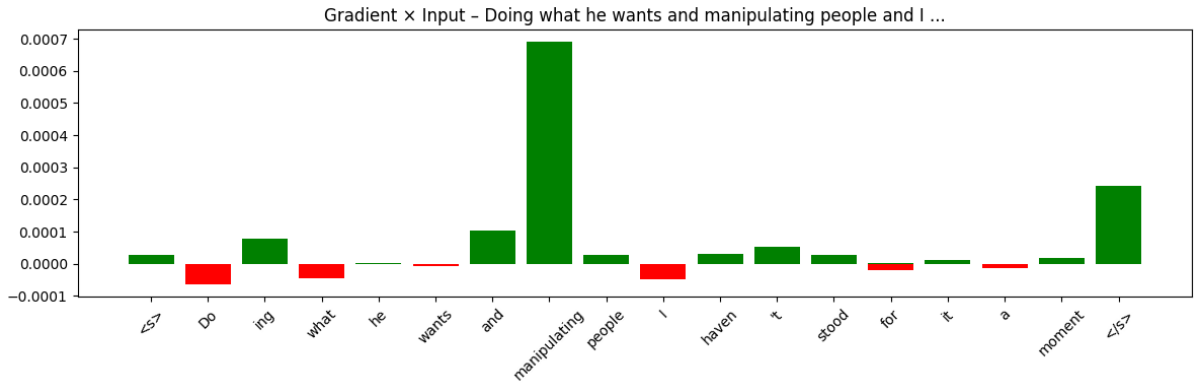


Fig 2 shows the GxI visualization for "doing what he wants and manipulating people and I haven't stood for it a moment"

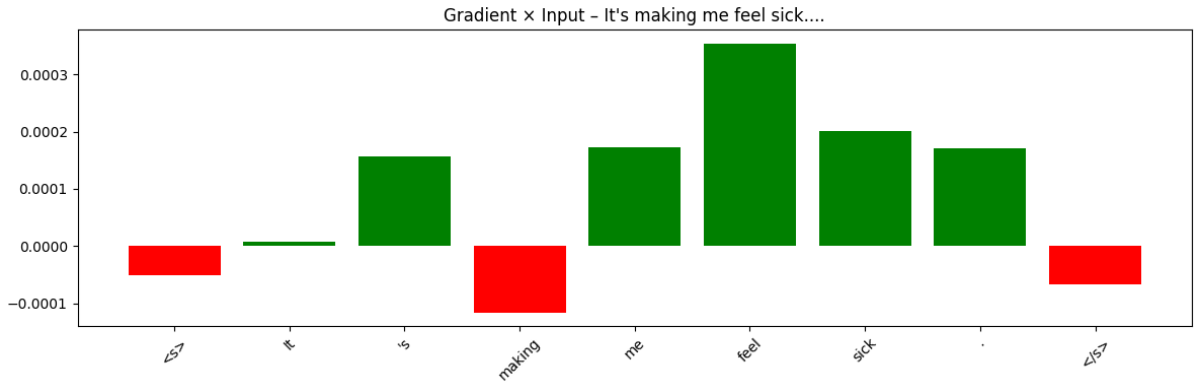


Fig 3: GxI visualization for “it’s making me feel sick”

However, the model sometimes assigns relevance to neutral tokens such as “him” or “get” in sentences such as “Don’t get angry at me”(fig 4) or “He told Jerry that I was against him trying to get him out”(fig 5) which shows that the model sometimes focuses more on sentence structure or frequent pronouns rather than genuine emotion clues.

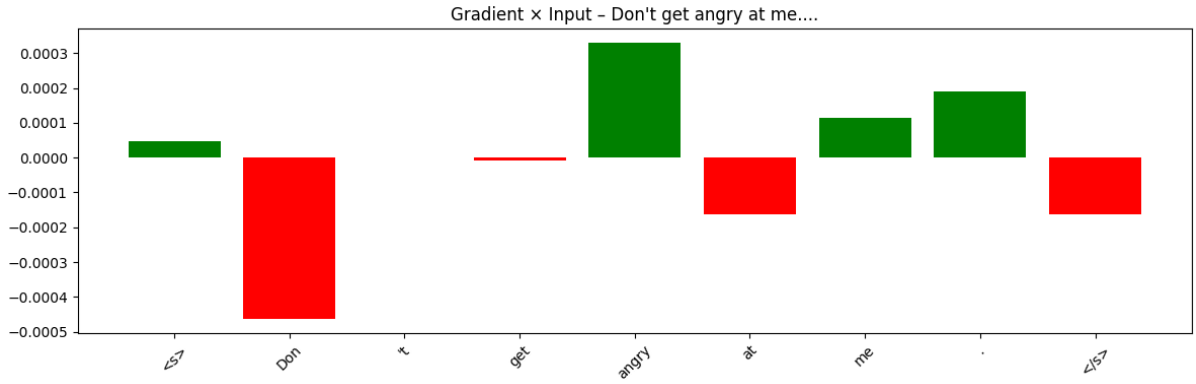


Fig 4: GxI bar plot of the sentence “Don’t get angry at me”

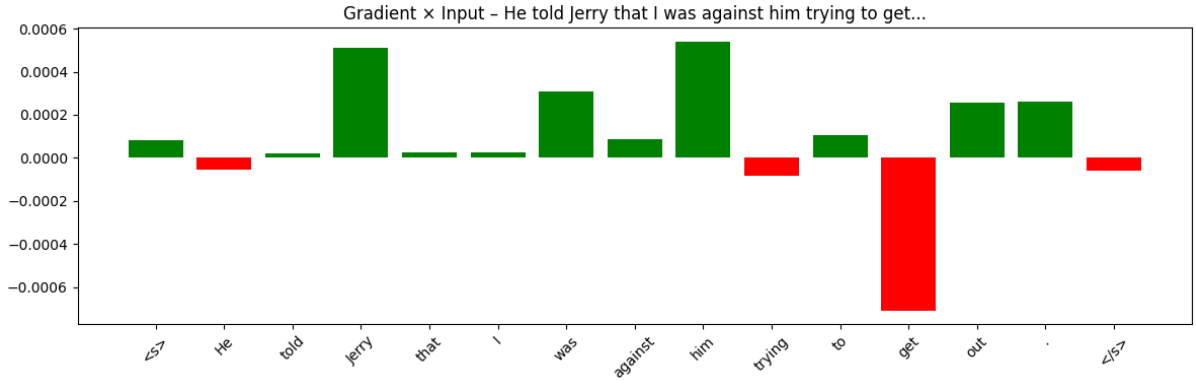


Fig 5: GxI bar plot of the sentence “He told Jerry that I was against him trying to get him out”

Overall, these plots show that while our model does pick up on emotional keywords, it sometimes pays attention to context words that have structural importance but limited emotional importance.

Layer-wise Relevance Propagation:

Unlike Gradient x Input, the layer-wise Relevance Propagation (LRP) visualizations draw a clearer, more understandable painting of what tokens the model pays more attention to. While GxI highlights whether a word increases or decreases confidence in the predicted emotion, LRP only focuses on how much each token contributes, regardless of direction. This makes the output easier to interpret, especially in longer sentences that might have multiple emotional cues.

For example, in the sentence “It’s making me feel sick”, as can be seen in figure 6, LRP shows “sick” as one of the highest relevance tokens but surrounding neutral tokens such as “me” or “feel” contribute minimally. This can also be seen in the sentence “I don’t even feed bad” as shown in figure 7, the strongest bars appear on “bad” and “feel”, which are both key words for emotion labelling.

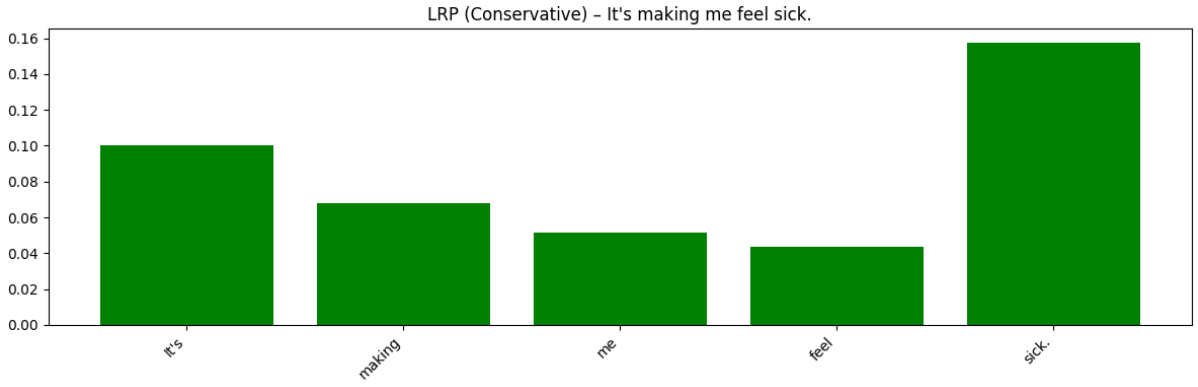


Figure 6 showing the LRP visualization of the sentence “it’s making me feel sick”

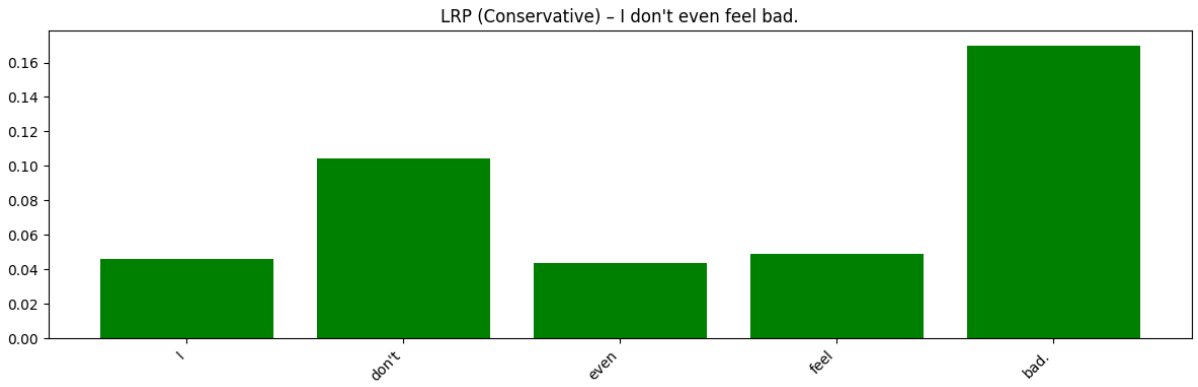


Figure 7 shows the LRP visualization of the sentence "I don’t even feel bad”

Another conclusion that can be taken from the visualization is that the model usually pays a good amount of attention to the last word in the sentence, which might suggest

that it is focusing on the ending sentence punctuation. This can be seen in the sentence “Have you gone and told anyone else?” since, as shown in figure 8, the model heavily focuses on “else?”. Suggesting that the model takes high interest in sentence ending punctuations such as ?, .,! etc.

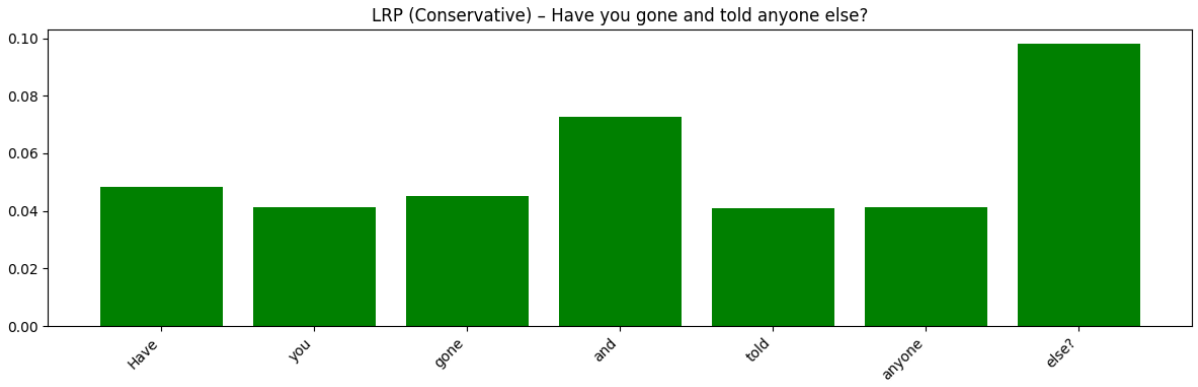


Figure 8 shows LRP visualization of the sentence “Have you gone and told anyone else?”

Overall LRP showed us that our models internal reasoning is more structured that what could be initially observed from GxI. The model mainly pays attention to emotionally meaningful words and sentence ending punctuations, the model still pays a good noticeable amount of attention to filler tokens but that is mostly when emotionally meaningful words are not present in the sentence.

Model Robustness with Input Perturbation:

The input-perturbation analysis builds directly upon the LRP results by testing how much the high relevance tokens matter to the model’s final confidence in its decision. In this procedure, tokens identified as least relevant by LARP are removed one by one, and the model’s prediction confidence is plotted as a function of the number of the words.

In most sentences, the model’s confidence declined gradually. This behavior indicated that RoBERTa-large does not exclusively rely on a single keyword but instead distributes its understanding across multiple contextually relevant tokens. It can also be seen that initially as the least relevant tokens are removed for example in the sentence “So there’s a chance that you could go home tonight” (figure 9), initially as the least relevant tokens are removed the models confidence increases which suggests that there are specific high importance tokens that the model focuses on which affect the models confidence, and in the later stages when those important tokens start get removed the model has a rapid decline (1 token removed) in its confidence.

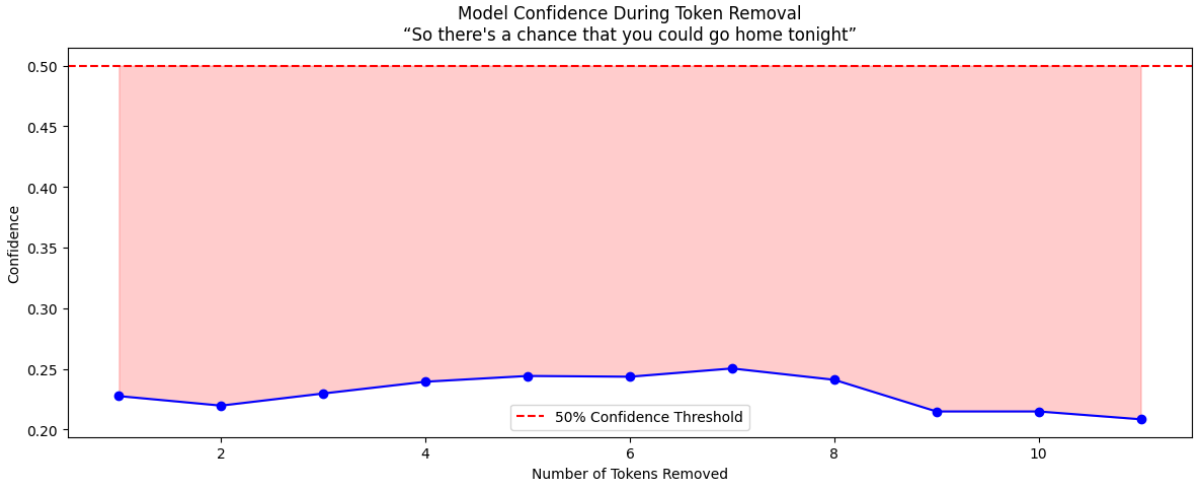


Figure 9 shows the confidence score of the model on the sentence “So there’s a chance that you could go home tonight” as the least relevant tokens are gradually removed.

Also, the overall low confidence levels that can be seen throughout the sentences, suggests that the mode is not very robust and lacks strong discriminative power between emotion categories. This means that while the model does pick up on emotional cues, it does so with uncertainty, which can be due to the fact that most of the times a lot of emotions are overlapping such as anger and disgust.